

TEACHER GUIDE

BBQ and Case • teacher guide

Teaching, assessment guidance and complete answers for the 50-page student edition

This guide accompanies a single 50-page student book: nine connected module readings, all 270 original multiple-choice questions and 810 options, all 27 written evidence tasks, twelve original folio prompts and their live-site evidence fields, four original drawing sheets and the current program's assigned supplementary learning. It does not set new construction specifications or assessment conditions.

How to use the guide

The next pages explain authority, evidence and the programme route. Knowledge-check answers then follow the original module and section order, ten questions per guide page, with the exact correct option and original rationale. Response guidance is keyed to the student page and stable field ID. A shared response area may meet related tasks only when it contains every named element; the source prompts remain individually visible.

Assess reasoning and genuine evidence

Multiple-choice items have source-defined answers. Written work and practical records may have more than one valid response. Use the guidance to assess source traceability, correct concepts, supported reasoning and the student's actual contribution. A polished account cannot replace a required observation, actual measurement, sample or teacher approval. Identify pending work honestly.

The student module-completion checkboxes and twelve Evidence checked and ready checkboxes record readiness only. They do not grant machine permission or automatically submit work. Optional authorised photographs remain optional unless a separate teacher-issued task requires photographic evidence. Where the original production-evidence folio prompt asks for photograph captions, the captions must relate to the student's own evidence.

The four landscape source-plan pages are part of the 50 pages. Print the PDF at actual size, preserve orientation and use the intended duplex setting. The other student content is portrait A4. Check one physical proof before a class run, especially page orientation and binding. Written source dimensions control; image scale does not supply missing measurements.

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Authority and boundaries

Use the current teacher-issued material for every practical decision

Drawings and the measured case base

The BBQ, case base, case lid and handle-components sheets form the original coordinated drawing set. The lid note requires the actual outside dimensions of the completed base to replace the nominal 510 and 330 values in the source calculation. Require a genuine measurement record, the correct outside faces and the teacher-confirmed source calculation. Do not impose a fixed lid size or invent a clearance, allowance, tolerance or hardware position.

Permission and process information

Current school SOPs, demonstrations, WMS, risk controls and teacher permission govern practical work. This book explains why controls and checks matter; it does not provide equipment settings or substitute a local procedure. Recheck permission when material, setup, product, drawing or conditions change. For finishes, distinguish the product label, technical information and SDS, and use their current authorised instructions.

Source discrepancies resolved in the edition

The website's week labels differ from the current 60-period teaching program, so this edition uses a module sequence without a fixed calendar. Source administrative statements about assessment weighting and timing conflict with draft or older materials. The teacher-issued current task remains authoritative; this edition omits those unconfirmed administrative claims while preserving all 270 source questions and all evidence prompts. No multiple-choice answer depends on the omitted dates or marks.

The supplemental program allocation is limited to the concepts and performance studios assigned to this project. It does not turn this textbook into the entire 200-hour course. The maintenance, ethical protocol, traditional/digital production, joining and two-technique finishing records require actual authorised evidence; a reading response alone does not establish practical performance.

Illustrations and authentic evidence

All new editorial images are illustrative. They support observation and explanation but establish no grade, project dimension, defect cause, permissible tolerance, product identity or finish suitability. They are not student work. Original source drawings are reproduced as complete sheets; native display contrast aids legibility without altering the source image bytes.

TEACHER GUIDE

Module 1 • checks 1–10

Reading the BBQ-and-case brief as a production contract

1. A. A production contract. The theory describes the approved project brief as a production contract because it defines required outcomes, limits and the evidence that must be collected.
2. B. A folio checklist. Translating the brief, drawings and evidence list into a folio checklist helps separate required actions from assumptions before production starts.
3. B. An identified function required by the brief. The theory states that a project requirement is a condition that must be met, such as an identified function, specified evidence or a teacher-controlled feature.
4. C. Confirm the detail with the teacher. The theory explains that unclear details should be confirmed with the teacher rather than being invented by the student.
5. B. It may be considered only when the brief allows flexibility. Preferences are only acceptable where the brief allows flexibility and cannot override drawings, required processes or approval points.
6. A. Identifying shared requirements, interfaces and evidence. The theory says the products should be planned together by identifying shared requirements, interfaces and evidence, not by guessing construction details.
7. B. Treat the feature as fixed unless the teacher approves a change. Authorised requirements remain fixed unless the teacher approves a change. Personal preferences cannot override compulsory project conditions.
8. B. By showing that production followed the approved brief. A checklist provides a record that practical work developed from the approved brief rather than from unverified personal interpretation.
9. C. Record allowable preferences separately from compulsory conditions. Keeping allowable preferences separate from compulsory conditions helps ensure required project expectations remain clear and traceable.
10. A. The teacher-issued project documents and demonstrations. The source boundary states that unresolved project-specific details are controlled by the teacher-issued project brief, drawings, SOPs, WMS template and demonstrations.

TEACHER GUIDE

Module 1 • checks 11–20

From hazard review to a task-specific WMS

11. A. To recognise what could cause harm. Hazard identification focuses on recognising sources of potential harm before work begins. It does not approve equipment use or replace operating procedures.
12. B. To provide the approved operating requirements for a particular tool, machine or process. An SOP defines the approved operating requirements for a specific tool, machine or process. It has a different purpose from a WMS.
13. A. Task sequence, hazards, controls, permission requirements and hold points. A WMS organises the planned work into stages and records the task, hazards, controls, permission requirements and hold points for each stage.
14. B. To avoid making hurried safety decisions during production. Planning ahead allows hazards, controls and checks to be considered before work starts instead of relying on rushed decisions during production.
15. A. Which different SOPs or demonstrations apply at different stages. The WMS links each stage of work with the relevant SOPs or demonstrations that apply to that part of the production process.
16. B. Because lathe operations are specifically identified as requiring a WMS. The theory states that lathe operations require particular attention because the authorised scope specifically requires a WMS for this work.
17. C. Evidence that planning has been completed. A completed WMS is evidence of planning. It does not authorise equipment use or replace demonstrations, supervision or approved procedures.
18. A. Hazard reviews identify risks, while SOPs define approved procedures. Hazard reviews focus on recognising risks, while SOPs specify the approved way to operate a particular tool, machine or process.
19. B. Stopping work at a planned point for checking or teacher authorisation. A hold point is a planned stage where work stops so required checks or teacher authorisation can occur before continuing.
20. C. The current teacher-issued SOPs, WMS template and demonstrations. The source boundary states that unresolved task-specific safety and operating details are controlled by the current teacher-issued SOPs, WMS template and demonstrations.

TEACHER GUIDE

Module 1 • checks 21–30

Inputs, processes and outputs as a planning model

21. B. Approved materials. Inputs are the resources needed before work begins, such as approved materials, drawings, information, equipment, time and teacher instructions. Completed components and folio evidence are outputs.

22. A. To transform inputs into products and evidence. Processes are the controlled transformations applied to inputs. They produce both physical work and evidence recorded throughout production.

23. C. The way stages connect through inputs, changes, checks and evidence. The theory explains that a production flowchart should show how stages connect, including what enters each stage, what changes, what is checked and what evidence is recorded, rather than being only a task list.

24. A. To pause production so quality or approval can be confirmed. A quality hold point deliberately pauses production so accuracy, fit, alignment, completeness or approval can be checked before work continues.

25. B. Return to an earlier stage for correction or further planning. A feedback loop directs the process back to an earlier stage so corrections, adjustments or additional planning can occur before production continues.

26. A. Approved materials, drawings and teacher instructions. Inputs include approved materials, drawings, measurements, information, equipment, time and teacher instructions. The other options list outputs.

27. B. Because it helps coordinate resources, sequence work, document quality and prevent unnoticed errors moving forward. The flowchart supports production by coordinating resources, sequencing work, documenting quality and reducing the chance of carrying errors into later stages.

28. C. Completed work or evidence produced during production. Outputs include completed components, assembled work, checked surfaces and folio evidence created throughout the production process.

29. A. The teacher-issued production plan, drawings and demonstrations. The source boundary states that the teacher-issued production plan, drawings and demonstrations control the exact sequence and any unresolved project-specific details.

30. C. What enters the process, what changes, what must be checked and what evidence should be recorded. Each stage should identify its inputs, the transformation taking place, the required quality checks and the evidence to record before progressing to the next stage.

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Module 2 • checks 1–10

Selecting stock form by function and manufacture

31. B. A form suited to function and manufacture. Choosing stock includes selecting a form that supports both the component's purpose and its manufacturing requirements.

32. A. The issued drawing and cutting information. The theory states that exact grades and sizes come from the issued drawing and cutting information.

33. C. Joining access. Joining access affects whether components can be assembled effectively using the approved manufacturing approach.

34. A. To reduce unnecessary waste through planning. Planning layouts and identifying usable offcuts improves efficiency without altering the authorised specification.

35. B. Reduce waste through careful planning. Efficiency comes from thoughtful planning, quantity checks and layout decisions rather than changing approved specifications.

36. C. Teacher-issued drawings, cutting information and demonstrations. The source boundary assigns unresolved stock specifications and project-specific details to teacher-issued project information.

37. A. Reassess the stock form because manufacture is affected. A suitable selection must support both function and practical manufacture, including adequate joining access.

38. B. Function, properties and manufacturing needs. The key takeaway emphasises balancing function, material properties and manufacturing requirements during selection.

39. B. It creates excessive waste. The theory explains that excessive waste can make an otherwise suitable stock form impractical.

40. C. To ensure the stock can be processed accurately. Available processes must be capable of accurately shaping, cutting, drilling or assembling the selected stock form.

TEACHER GUIDE

Module 2 • checks 11–20

Turning drawings into freehand sketches and a spreadsheet cutting list

- 41. B. Convert drawings into organised production information. Production information should be organised from verified drawings before any material is marked or cut.
- 42. A. Different views can reveal different component features. Cross-checking views combines their information and reduces the risk of overlooking an important feature.
- 43. C. Leave it unresolved until clarified. Missing data must not be invented or measured from the image; teacher clarification controls unresolved information.
- 44. B. Show how the component has been interpreted. A freehand sketch demonstrates understanding of general form, important features and relationships between drawing views.
- 45. A. Brief annotations linked to verified dimensions or notes. Brief annotations can identify where confirmed drawing dimensions or notes apply without replacing the formal drawing.
- 46. C. An unconfirmed material allowance. Unconfirmed quantities, allowances and tolerances must remain blank until authorised information or teacher clarification is provided.
- 47. B. Using consistent units for every entry. Consistent units help entries match written drawing information and reduce avoidable interpretation errors during production.
- 48. A. The current revision or issue date. Recording the supplied revision or issue date helps ensure the cutting list uses the current authorised information.
- 49. C. When an authorised drawing change occurs. The spreadsheet must reflect authorised changes so outdated information does not guide later production decisions.
- 50. B. Cross-check drawings, sketches, cutting list and teacher clarification. Cross-checking all verified sources prevents an early information error from becoming wasted material during production.

TEACHER GUIDE

Module 2 • checks 21–30

Datums, orientation and quality gates for rails and end caps

- 51. B. An agreed reference edge, face or point. Using one agreed reference helps measurements stay consistent and reduces cumulative errors across related components.
- 52. A. To limit cumulative marking error. Small marking differences can accumulate, so a consistent datum helps related parts align more accurately later.
- 53. B. Matching parts. Orientation marks help keep matching components correctly paired and facing their intended direction throughout production.
- 54. C. To reduce the chance of processing the wrong side. Clearly identifying the waste side helps prevent removing material from the incorrect side of a marked line.
- 55. A. Before irreversible work. Inspection before irreversible work allows corrections while marks and component positions can still be changed.
- 56. A. The component with the drawing. Comparing the component with the drawing helps confirm accuracy before permanent processing begins.
- 57. B. It prevents one early error being repeated. Checking accuracy throughout production helps detect problems before they affect multiple related components.
- 58. B. Orientation and relationship to matching parts. Inspection should confirm orientation and matching relationships before processing makes changes difficult or impossible.
- 59. C. Transferring measurements from one consistent datum. A single consistent datum reduces the build-up of small measurement differences across multiple components.
- 60. A. The teacher-issued drawing, production plan and demonstration. Project-specific details come only from the approved drawing, production plan and teacher demonstration, not assumptions.

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Module 3 • checks 1–10

Hole location is a design feature, not just a mark

61. A. Its location affects fit, alignment and function. Hole position influences how components fit and function, so its location must follow verified design information.

62. B. The teacher-issued drawing. Hole locations should always come from verified drawing information rather than estimates or copied parts.

63. A. The correct drawing view, written information and component identity. Checking the drawing view, written information and component identity helps prevent drilling the wrong feature or part.

64. C. To provide the reference for centre locations. Datums provide a consistent reference so measurements remain accurate and repeatable during marking out.

65. A. Small errors can shift the feature. Changing reference edges allow small errors to accumulate, affecting later alignment and assembly.

66. B. They confirm the correct face, side or end is being processed. Orientation marks reduce the chance of drilling an accurately marked feature into the wrong component face.

67. A. Its design purpose. Understanding the design purpose helps ensure the hole is marked and checked correctly before drilling.

68. A. Compare the workpiece with the drawing, verify the datum, confirm the marked centre and check the part label. A complete verification confirms the feature is correctly planned before drilling permanently changes the workpiece.

69. B. To show it was traced to verified design information. Being able to explain the source shows the location was based on verified information rather than assumption.

70. B. It provides evidence that accuracy was planned and controlled. Recording the check documents that accuracy was verified before the irreversible drilling process began.

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Module 3 • checks 11–20

Drill-press principles behind the teacher-issued SOP

71. A. The operator, workpiece and planned task are ready under required supervision. Authorisation confirms readiness and supervision before work begins, rather than replacing the approved operating procedure.

72. B. To prevent the component moving or rotating during drilling. Workholding keeps the component stable so drilling forces do not cause unwanted movement or rotation.

73. C. They help maintain separation from moving parts and avoid unwanted contact. Guarding and suitable clearance support safe operation by preventing unintended contact with moving parts or the setup.

74. A. To manage the cutting action instead of forcing it. Controlled feed means the cutting action is managed deliberately rather than applying unnecessary force.

75. B. It can affect visibility, cutting quality and inspection. Swarf can influence how well the work can be seen, assessed and cut throughout the drilling process.

76. C. The current SOP and teacher demonstration. The theory states that exact operating methods are controlled by the current authorised procedure and demonstration.

77. A. As a deliberate stage before inspection. Stopping is part of the planned process and prepares the machine for safe inspection of the work and result.

78. B. Whether the drilled feature matches the approved drawing and suits the next stage. Inspection verifies both the drilled feature and the component's readiness for the following production stage.

79. B. Authorisation, supervision and the demonstrated procedure. Understanding the concepts supports safe practice but never replaces authorised procedures or required supervision.

80. C. The current SOP rather than memory or general knowledge. Current authorised procedures take priority because they provide the approved method for the specific task being completed.

TEACHER GUIDE

Module 3 • checks 21–30

Inspecting drilled features and responding to faults

81. A. The completed feature against the approved drawing or quality criterion. Inspection compares completed work with approved requirements rather than judging appearance or comparing with other components.
82. B. It is suitable for later fitting or assembly. Suitability for later fitting or assembly is confirmed by comparison with approved requirements, not visual similarity alone.
83. A. Recorded measurements. Recorded measurements provide objective evidence that can be compared with approved drawings or quality criteria.
84. B. A fault category. Fault categories describe what was observed without proving why the problem occurred.
85. A. Investigate the likely cause using available evidence. Evidence should be reviewed before proposing corrections because similar faults can have different causes.
86. A. The demonstrated process. Reviewing the demonstrated process can help identify likely causes without relying on unsupported assumptions.
87. A. Because several causes may produce a similar fault. Different causes can create similar outcomes, so evidence is needed before reaching conclusions.
88. B. The feature should be recorded and referred to the teacher. Recording the fault and obtaining teacher review preserves evidence before any authorised rework is considered.
89. A. It may remove evidence or worsen the fault. Unauthorised changes may destroy useful evidence, worsen defects or affect later production stages.
90. A. Quality was controlled throughout production. Careful inspection shows quality was managed throughout production rather than judged only at the end.

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Module 4 • checks 1–10

Reading a lathe operation through the WMS

91. A. Connect the drawing requirement to the approved practical process. A WMS links the drawing, approved process and workpiece to maintain traceability throughout the operation.

92. B. The teacher-approved setup. The approved setup is identified before machining so the planned process matches authorised conditions and requirements.

93. C. To check work before irreversible stages continue. Hold points provide deliberate checks so early errors are identified before they affect later machining stages.

94. A. Planned tasks, hazards, controls, permissions and hold points. Controlled stages organise essential planning information so each part of the operation can be reviewed appropriately.

95. B. The demonstrated process, current setup and present conditions. Approval depends on the demonstrated process, current setup and existing conditions rather than appearance or elapsed time.

96. C. Stop and report the change. Changes to approved conditions require work to stop so the WMS and authorisation can be reviewed.

97. A. Linking the drawing to the completed work through the approved process. Traceability links the drawing, approved process and completed feature instead of relying on memory or assumptions.

98. B. When the workpiece, drawing, equipment, setup or conditions change. The WMS should be reviewed whenever approved conditions change rather than assumed to remain automatically valid.

99. C. The current lathe SOP. The WMS supports planning and evidence but does not replace the current SOP or demonstrated procedures.

100. A. Use hold points for inspection. Inspection hold points help confirm the workpiece or setup before irreversible machining continues to the next stage.

TEACHER GUIDE

Module 4 • checks 11–20

Cutting action, reference surfaces and dimensional control

101. A. To preserve shape, dimensions and surface condition. Progressive removal allows controlled decisions before each change permanently affects the developing component and its accuracy.

102. B. Reference surfaces. Reference surfaces help keep later measurements and decisions connected to the same verified starting point throughout machining.

103. C. Cumulative error and difficult measurement comparison. Changing references can introduce cumulative error, making later measurements harder to compare with the approved drawing.

104. A. To compare the developing component with verified drawing information. Hold points allow deliberate checks before more material is removed, helping identify problems while correction may still be possible.

105. B. Removed material cannot simply be replaced. Early checks reduce the chance of irreversible errors because removed material cannot simply be restored later.

106. C. Verified measurements, the approved drawing and teacher-controlled criteria. Dimensional acceptance depends on verified evidence rather than visual impressions or production speed.

107. A. A planned transformation. Machining is presented as a controlled process where each step is guided by planned decisions and verification.

108. B. To provide evidence that quality was deliberately managed. Recorded checks demonstrate that quality was monitored systematically throughout machining rather than assumed afterwards.

109. C. Appearance can support inspection but cannot confirm accuracy alone. Visual inspection is useful, but dimensional accuracy still requires verified measurements against approved requirements.

110. A. The established datum or reference surface. Maintaining the same reference surface supports consistent measurements and reduces the likelihood of cumulative error.

TEACHER GUIDE

Module 4 • checks 21–30

Surface quality, fit and documented teacher review

111. B. Size, form, surface condition and functional fit. Quality depends on considering size, form, surface condition and functional fit together rather than any single characteristic.

112. A. A smooth-looking feature may still have incorrect size, form or function. A good-looking surface does not guarantee the feature meets dimensional, form or functional requirements.

113. C. Approved measurements, observations, photographs or annotations. Inspection records should use approved evidence that documents the observed quality of the machined feature.

114. B. Identify it honestly. Honest identification supports traceable quality decisions and prevents problems being concealed.

115. A. To stop it being accidentally used later. Separating doubtful components reduces the risk of unsuitable work entering later production stages.

116. C. Teacher review. Teacher review ensures acceptance or correction follows approved quality processes rather than assumptions.

117. B. It may remove evidence, alter another feature or worsen the problem. Unplanned rework can damage evidence or create additional faults that complicate quality assessment.

118. A. To create traceable evidence of controlled quality decisions. Documentation provides a record showing quality decisions were based on controlled processes rather than guesses.

119. B. Whether the feature can perform its intended role within the developing assembly. Functional fit considers whether the machined feature is suitable for its intended purpose in the assembly.

120. C. The teacher-issued drawing, acceptance criteria, machining SOP and demonstration. Project-specific inspection, fit and rework requirements are controlled by the approved documents and teacher demonstration.

TEACHER GUIDE

Module 5 • checks 1–10

Planning plate work from verified references

121. A. Verified drawing information. Verified drawing information provides the authorised reference for identification, layout and later production decisions.

122. A. To reduce cumulative error. A consistent datum keeps measurements linked to the same reference and reduces errors building across the layout.

123. B. To show how the plate relates to surrounding components. Orientation marks help maintain the correct relationship between the plate and associated components throughout production.

124. B. The marking against the drawing. Checking marked information against the drawing helps identify errors before they become difficult or impossible to correct.

125. A. Verified dimensions or specifications. Material economy should reduce waste without changing any authorised dimensions or specifications from verified references.

126. B. To inspect identity, orientation, marking and condition. Quality hold points provide planned opportunities to verify important production information before work continues.

127. A. The teacher-issued plan. The teacher-issued plan controls unresolved sequencing and methods, ensuring production follows the approved process.

128. B. Using clear component identification and recorded checks. Component identification and recorded checks link the workpiece back to verified drawing information through each stage.

129. A. Usable areas, existing edges and other components' needs. Planning stock use considers available material efficiently while preserving every authorised requirement from the drawing.

130. B. Any unresolved drawing details. Unresolved drawing details should be confirmed first so planning and marking remain based on verified information rather than assumptions.

TEACHER GUIDE

Module 5 • checks 11–20

Grills and guard: function, spacing and manufacturability

131. A. Because their function depends on consistency and alignment. Related components should be assessed together because consistency and alignment influence their intended function and overall quality.

132. B. The teacher-issued drawing. Spacing and alignment should be verified against the approved drawing rather than estimated from appearance or another part.

133. A. To reduce repeated differences across related components. Consistent references reduce accumulated variation, helping related components remain aligned and function as intended.

134. C. Use verified references from the drawing. Appearance alone is not sufficient because verified references provide evidence that spacing matches the approved information.

135. B. Considering access for processing, inspection and maintenance. Manufacturability considers practical access throughout production, inspection and later maintenance, not only accurate fabrication.

136. A. They can affect overall fit, performance and presentation. Small differences can accumulate across related parts, influencing function, fit and the finished appearance.

137. B. Before assembly makes correction more difficult. Checking related parts before assembly helps identify and correct inconsistencies while changes are still practical.

138. C. Photographs, observations and verified checks. Evidence gathered during production shows consistency was verified rather than judged only from the finished result.

139. A. The feature may be difficult to manufacture or inspect. Manufacturability includes practical access, so restricted inspection or processing access may make a design unsuitable.

140. B. The teacher-issued drawings, approved quality criteria and demonstrations. Project-specific spacing and manufacturing details are controlled by approved drawings, quality criteria and teacher demonstrations.

TEACHER GUIDE

Module 5 • checks 21–30

Fit-up before permanent joining

141. B. Diagnose problems while parts can still be separated. Fit-up identifies issues before permanent joining, allowing components to be adjusted, repositioned or reviewed while changes remain practical.

142. A. Orientation, contact, alignment, access and assembly sequence. Fit-up examines how related components work together, not unrelated workshop or presentation details.

143. B. It may reveal that one correctly fitting part blocks later assembly. A component may fit individually yet prevent correct positioning or access during later assembly stages.

144. A. They provide evidence that component relationships were examined before joining. Photographs and annotations document that fit-up checks were completed before permanent joining began.

145. C. Mask poor contact or introduce movement when released. Restraint can hide problems or allow movement after release, so fit-up must be evaluated carefully.

146. A. Uneven heating and cooling may affect alignment or form. The theory explains that heating and cooling can change alignment or form and should be anticipated conceptually.

147. B. Report it before permanent joining. Problems or uncertainty should be reported before joining makes corrections more difficult or irreversible.

148. B. Verified references including the teacher-issued drawing. Fit-up decisions are checked against verified references rather than appearance or unverified examples.

149. A. Appearance alone does not confirm compliance with verified references. Visual appearance alone cannot confirm correct fit, alignment or compliance with approved references.

150. C. The teacher-issued drawing, approved joining method and demonstration. Project-specific fit-up and joining requirements come from the authorised drawing, method and teacher demonstration.

TEACHER GUIDE

Module 6 • checks 1–10

Stability begins with geometry and matched components

151. A. Accurate geometry. Stable support is achieved through accurate geometry before final assembly, not by relying on later adjustment.

152. B. To prevent accumulated variation. Using consistent reference points helps prevent small differences accumulating and affecting the completed assembly.

153. C. To preserve the intended relationship between components. Orientation marks help matching faces, edges or ends remain correctly related during assembly and inspection.

154. A. Related components and their geometry. Comparing related components against verified references helps identify problems before permanent assembly.

155. B. Nothing about the exact cause. Visible symptoms alone do not identify the underlying cause and require further inspection and measurement.

156. B. Incorrect orientation. Incorrect orientation is one possible cause, alongside inconsistent references, matched-component differences and accumulated assembly error.

157. C. Record the symptom and inspect the assembly. Recording the symptom before inspection supports evidence-based diagnosis instead of guessing or making unnecessary changes.

158. A. Verified measurements. Verified measurements provide reliable evidence for locating variation instead of relying on assumptions.

159. B. Teacher review. Teacher review helps ensure corrections are appropriate and protects component quality from unnecessary changes.

160. C. Controlled geometry. The process demonstrates that stability comes from controlled geometry supported by measurement, inspection and review, not improvised adjustment.

TEACHER GUIDE

Module 6 • checks 11–20

Assembly order, access and hold points

161. A. It controls dependencies between components and processes. Assembly order manages dependencies that affect access, alignment, inspection and later production stages.

162. B. What must already be complete and what access must remain. Checking prerequisites and remaining access helps prevent later assembly problems and supports controlled production.

163. C. Restrict later positioning, inspection or processing. A correctly fitted component can still create later difficulties if it blocks access or inspection.

164. A. Pause production for verified checks before continuing. Hold points verify important features before errors become harder to correct in later assembly.

165. B. Component identity, orientation, contact, alignment and required evidence. Hold points compare verified features and evidence against approved requirements before production continues.

166. C. An unnoticed error may affect several completed components. Missing a hold point allows undetected errors to carry into later work, increasing correction effort.

167. A. The production flowchart and folio record. Updating production records maintains traceability and documents the controlled change to the approved sequence.

168. B. Because later assembly may reduce access to important areas. Some features become difficult to inspect or finish once additional components are assembled.

169. C. Recording the reason, approval and effect of the change. Traceability depends on documenting why the sequence changed, who approved it and its production impact.

170. A. The teacher-issued assembly plan, drawings and demonstrations. The supplied source boundary states that unresolved production details come from approved teacher documentation and demonstrations.

TEACHER GUIDE

Module 6 • checks 21–30

Quality assurance during structural assembly

171. A. Checking results at particular stages. Quality control focuses on checking results at specific stages before work progresses to the next step.

172. B. A system showing quality was planned, checked and maintained. Quality assurance links planning, checking, recording and maintaining quality across the whole project, not just one inspection.

173. C. Photographs with annotations linked to checks. Traceable photographs and annotations explain what was inspected, the evidence found and any required follow-up.

174. A. The relevant component, drawing requirement, production stage and teacher-issued criterion. Traceable records link observations to approved references so inspection evidence can be followed through the project.

175. B. As a non-conformance. A possible or confirmed issue should be recorded as a non-conformance rather than ignored or exaggerated.

176. C. The teacher should review it. Assembly should not continue past a possible non-conformance until the teacher has reviewed the affected work.

177. A. Claiming an unsupported cause. Reports should record observed evidence without hiding issues or making unsupported claims about their cause.

178. B. Teacher authorisation. Rework must be authorised by the teacher before it proceeds and should then be documented.

179. C. The issue, approved response and later inspection result. A complete rework record shows the issue, approved action and evidence from the later inspection.

180. A. Assembly quality was built through controlled decisions. Documented rework demonstrates that quality resulted from controlled, traceable decisions rather than appearance alone.

TEACHER GUIDE

Module 7 • checks 1–10

Surface condition controls finish performance

181. A. It determines whether the finish can bond consistently. A clean, consistently prepared surface supports bonding, appearance and protection before the specified finish is applied.

182. B. Poor surface preparation. Extra product cannot overcome burrs, contamination, scratches or uneven preparation that should have been addressed beforehand.

183. A. Preparation. Preparation creates the required surface condition before the specified finish is applied using the authorised method.

184. C. The finish may not yet be fully cured. Drying and curing are different stages, so a dry surface may still be developing its intended properties.

185. A. To identify surface issues requiring review. Hold-point inspections help detect burrs, contamination, inconsistent texture or damage before the finish is applied.

186. B. Photographs and annotated observations. Photographs with annotated observations provide traceable evidence that preparation was completed systematically before application.

187. C. Contamination remaining on the surface. Contamination can reduce adhesion and protection, making careful preparation essential before applying the specified finish.

188. A. Current product information and teacher instructions. Current product information and teacher instructions control unresolved details throughout the finishing process.

189. B. Application places the specified finish onto the prepared surface. Preparation creates the surface, while application places the specified finish onto that prepared surface.

190. C. They can affect appearance, adhesion and protection. Following authorised instructions helps maintain the intended appearance, adhesion and protective performance of the finish.

TEACHER GUIDE

Module 7 • checks 11–20

Selecting and applying a finish from authoritative information

191. A. Confirmed performance criteria. A suitable finish is justified by confirmed performance criteria, not appearance or other unrelated preferences alone.

192. B. Future maintenance requirements. Future maintenance is one of several confirmed factors considered alongside protection, appearance and intended use conditions.

193. C. The Safety Data Sheet (SDS). The SDS provides authoritative information about hazards, handling, storage and other controls for the product.

194. A. Check reliable product information. Suitability claims should be supported by reliable product information rather than assumptions or past experience.

195. B. Identify intended use and approved application requirements. The product label identifies intended use and approved application requirements, while hazard information is detailed in the SDS.

196. C. Stop and seek clarification. Differences from the approved method require students to stop and seek clarification before continuing.

197. A. Label, SDS and technical information. Current labels, SDS and technical information provide the evidence needed to justify finish selection.

198. B. Why it meets the confirmed criteria. Folio evidence should explain how the specified finish satisfies the confirmed performance and maintenance criteria.

199. C. Teacher-specified product information and demonstration. The current teacher-specified product information and demonstration control unresolved handling, application, suitability and curing details.

200. A. To create a traceable link between planning, protection and the completed result. Consulting current instructions supports traceability from production planning through material protection to the completed result.

TEACHER GUIDE

Module 7 • checks 21–30

Finish inspection, maintenance and environmental responsibility

201. A. Teacher-approved criteria. Inspection should compare the finish with approved criteria, producing evidence rather than relying on overall appearance.

202. B. Annotated observations identifying inspected areas. Annotated observations identify what was inspected and clearly record any uncertainty needing teacher review.

203. C. They may reveal defects or inconsistent coverage. Edges and corners can show defects or uneven coverage that affect comparison with approved quality criteria.

204. A. To prevent damage before intended properties develop. Protection during drying and curing helps prevent damage before the finish develops its intended properties.

205. B. Current product information and teacher directions. Curing conditions must always follow current product information together with teacher directions, not assumptions.

206. C. Maintenance, repairability and expected service life. A sound evaluation considers ongoing maintenance, repairability and expected service life using confirmed information and evidence.

207. A. It may require more maintenance or earlier replacement. Frequent maintenance or early replacement can increase material use and ongoing resource demands over time.

208. B. According to teacher directions and authorised school procedures. Waste management should follow authorised school procedures and teacher directions to support responsible practice.

209. C. Observed evidence supported by confirmed product information. Observed evidence and confirmed product information provide a reliable basis for evaluating finish quality and performance.

210. A. What was inspected and any uncertainty requiring review. Inspection photographs should show inspected areas and clearly identify any uncertainty requiring teacher review.

TEACHER GUIDE

Module 8 • checks 1–10

From three-dimensional case to flat development

211. A. A flat representation of surfaces forming a three-dimensional case. A development shows how flat surfaces relate before forming, helping students understand the completed three-dimensional case.

212. B. Because development lines may represent different features or production stages. Development lines can represent different features or stages, so they must be interpreted alongside the approved case drawings.

213. A. They establish where information is located and transferred. Reference lines organise and transfer information accurately between the development and the completed case.

214. B. A correctly shaped area may be formed in the wrong direction. Orientation marks help keep faces and edges correctly related, preventing correctly shaped areas from ending up incorrectly positioned.

215. C. The approved drawing, supplied formula and teacher demonstration. Allowance values are controlled by approved information rather than estimates or visual guesses.

216. A. To account for the material and geometry during transformation. Allowances account for how flat material changes when transformed into the required three-dimensional form.

217. B. To confirm verified information before irreversible processing. Hold points provide opportunities to verify important information before later operations make correction more difficult.

218. C. As connected surfaces that relate to the completed case. A development should be interpreted as connected surfaces that together create the finished three-dimensional case.

219. A. Some features may no longer remain visible or accessible. Later forming can reduce access or visibility, making important checks and corrections more difficult to complete.

220. B. The teacher-issued drawings, approved formulas, SOPs and demonstrations. The source boundary requires unresolved details to be controlled by approved documents and teacher guidance, not assumptions.

TEACHER GUIDE

Module 8 • checks 11–20

Bending sequence, accumulated error and case fit

221. B. Early bends affect later geometry, access and alignment. Following the approved sequence helps maintain geometry, access and alignment as each bend influences later forming stages.

222. A. To reduce accumulated error when comparing with the drawing. Consistent datums provide stable references, reducing the chance that small differences build into larger alignment problems.

223. C. Small differences may accumulate across the case. Changing reference points can allow small variations to build until noticeable fit or geometry problems appear.

224. A. Orientation, reference lines, developing geometry and trial fit. Hold points help identify variation early by checking orientation, references, geometry and trial fit before irreversible work continues.

225. B. Misaligned mating edges or uneven case geometry. Accumulated error often appears as overall fit or alignment problems rather than one obvious large mistake.

226. C. Partial recovery towards the original shape after forming force is removed. Springback describes a material's tendency to recover partly after forming, although its significance depends on the approved process.

227. A. Document it and refer it to the teacher before continuing. Unexpected movement or mismatch should be recorded and referred rather than corrected through unapproved adjustments or guesses.

228. B. To identify where variation first appeared. Intermediate checks help locate the first appearance of variation before further forming makes correction more difficult.

229. C. Guessing compensation values. The theory states that compensation values must not be guessed because relevance depends on the confirmed material and approved process.

230. A. Each bend can influence later features and references. Each forming stage influences later geometry, reference points and alignment, making the approved sequence important throughout the process.

TEACHER GUIDE

Module 8 • checks 21–30

Hinges and handle as an evidence-based design problem

231. A. Define confirmed constraints. Confirmed constraints from authorised sources establish what the design must resolve before alternatives are generated.

232. B. As a question requiring confirmation. Unverified assumptions should remain recorded questions until confirmed through teacher-issued information or authorised review.

233. C. To compare alternatives before committing. Generating alternatives allows students to evaluate different solutions against approved functional and manufacturing criteria.

234. A. Movement, access and ergonomics. Movement, access and ergonomics are identified criteria for comparing how effectively each arrangement may function.

235. B. To make design reasoning visible. Tables, sketches and design notes show how alternatives were compared and why one may be preferred.

236. C. It can reveal practical difficulties. A prototype may expose restricted movement, poor access, awkward handling or manufacturing problems hidden in sketches.

237. A. Results, problems and unresolved questions. Honest documentation must include trial results, identified problems and any questions that remain unresolved.

238. B. Approved criteria and testing results. Selection should be justified through approved comparison criteria and evidence gathered from any authorised testing.

239. C. After teacher review and required testing. Such claims require teacher review and completion of the required testing, not student judgement alone.

240. A. A controlled design process. These records demonstrate that the final decision developed systematically through comparison, testing and authorised approval.

TEACHER GUIDE

Module 9 • checks 1–10

PMI evaluation that moves beyond opinion

241. B. The teacher-issued brief and approved criteria. A PMI evaluation is evidence-based and compares the project with the approved expectations rather than personal opinion.

242. A. Evidence such as photographs, measurements, testing records or folio notes. A Plus is strengthened by evidence that demonstrates the successful feature or process outcome being described.

243. B. The effect of the problem. Describing the effect first separates observed evidence from later explanations about why the issue may have occurred.

244. C. Only when supported by available evidence. The theory requires causes to remain tentative unless evidence is sufficient to support the conclusion.

245. A. It identifies a meaningful result or relationship worth further consideration. Interesting observations extend evaluation by exploring meaningful findings rather than repeating praise or opinions.

246. B. Interesting. Unexpected relationships between production decisions and outcomes are identified as Interesting observations in the PMI.

247. C. One achievable improvement based on evidence. The final improvement should respond to evidence, address a likely cause and remain realistic within project constraints.

248. A. Link each observation to approved criteria and evidence. PMI observations gain credibility when they are clearly connected to approved criteria and supporting evidence.

249. B. So the recommendation can realistically be applied. Useful improvements are achievable and grounded in the project's available processes, evidence and constraints.

250. B. The teacher-issued brief, approved criteria and completed project evidence. Where uncertainty remains, the brief, approved criteria and completed project evidence determine justified evaluation claims.

TEACHER GUIDE

Module 9 • checks 11–20

Building the project report as an evidence chain

251. B. Form an evidence chain from brief to evaluation. The report should connect decisions, production evidence and evaluation into a traceable sequence rather than disconnected documents.

252. A. The approved brief. The evidence chain starts with the approved brief before later planning, production and evaluation evidence is presented.

253. C. To show stages, hold points, problems and authorised responses in sequence. Chronological order helps readers follow how production progressed and how important decisions affected outcomes.

254. A. A concise caption explaining its purpose. Captions should identify what is shown, why it matters and which requirement or decision the evidence supports.

255. B. Record the as-built result and acknowledge the change. An accurate report records what was actually made and openly acknowledges changes from earlier planning.

256. A. Teacher-issued drawings, product information and demonstrations. Identifying sources shows where important production decisions were informed by authoritative information.

257. C. Do not invent it or recreate it inaccurately. Missing evidence must remain honest rather than being invented or reconstructed inaccurately.

258. A. Teacher to confirm. Unverified administrative details should remain clearly marked as needing teacher confirmation before being treated as facts.

259. B. Brief, production, evaluation. The report should logically progress from the brief through production evidence to the final evaluation.

260. C. The current Task 1 handbook, teacher-issued brief, drawings and submission directions. When information is unresolved, authoritative task documents determine the correct requirements rather than assumptions.

TEACHER GUIDE

Module 9 • checks 21–30

Final quality assurance and authentic submission

261. B. Check both the finished project and its supporting evidence. Final quality assurance verifies both the physical project and the folio accurately and consistently represent the completed work.

262. A. What was actually produced. The folio should match the as-built result rather than an earlier plan or an idealised version.

263. C. Identify the approved change. Approved changes should be identified so the evidence accurately explains how the final result developed.

264. A. To support an accurate evidence chain. Consistent evidence helps demonstrate that the folio truthfully documents the completed project from multiple sources.

265. B. Report it honestly. Honest reporting maintains the integrity of the submission even when problems remain unresolved.

266. A. After an authorised confirmation. Inspection claims must only be made after confirmation through the authorised process described by project requirements.

267. B. To recognise sources that influenced decisions. Acknowledging sources makes it clear where important decisions or information came from and supports authentic work.

268. C. Acknowledged honestly. Assistance should always be acknowledged rather than presented as original student work.

269. A. Its completeness, readability and sequence. Evidence should be complete, readable and correctly ordered so it can be followed and verified.

270. B. The teacher-issued brief, drawings, criteria, handbook and submission directions. When details remain unresolved, the authorised project documents determine the correct inspection and submission requirements.

TEACHER GUIDE

Student page 2 · response guidance · 1/2

Your learning and evidence route

Module 1 completion [m01-complete]. Tick this module only after its theory, original checks and written evidence have been completed and saved or printed as directed. The checkbox does not grant practical permission or submit the folio. Pending work remains unticked.

Module 2 completion [m02-complete]. Tick this module only after its theory, original checks and written evidence have been completed and saved or printed as directed. The checkbox does not grant practical permission or submit the folio. Pending work remains unticked.

Module 3 completion [m03-complete]. Tick this module only after its theory, original checks and written evidence have been completed and saved or printed as directed. The checkbox does not grant practical permission or submit the folio. Pending work remains unticked.

Module 4 completion [m04-complete]. Tick this module only after its theory, original checks and written evidence have been completed and saved or printed as directed. The checkbox does not grant practical permission or submit the folio. Pending work remains unticked.

Module 5 completion [m05-complete]. Tick this module only after its theory, original checks and written evidence have been completed and saved or printed as directed. The checkbox does not grant practical permission or submit the folio. Pending work remains unticked.

Module 6 completion [m06-complete]. Tick this module only after its theory, original checks and written evidence have been completed and saved or printed as directed. The checkbox does not grant practical permission or submit the folio. Pending work remains unticked.

Module 7 completion [m07-complete]. Tick this module only after its theory, original checks and written evidence have been completed and saved or printed as directed. The checkbox does not grant practical permission or submit the folio. Pending work remains unticked.

TEACHER GUIDE

Student page 2 · response guidance · 2/2

Your learning and evidence route

Module 8 completion [m08-complete]. Tick this module only after its theory, original checks and written evidence have been completed and saved or printed as directed. The checkbox does not grant practical permission or submit the folio. Pending work remains unticked.

Module 9 completion [m09-complete]. Tick this module only after its theory, original checks and written evidence have been completed and saved or printed as directed. The checkbox does not grant practical permission or submit the folio. Pending work remains unticked.

TEACHER GUIDE

Student page 28 · response guidance

Develop, form and check the case

Source task 8.2 [m08-task02]. A forming hold point preserves evidence before the next stage. Compare verified marks with their datum/drawing, check actual setup and orientation under teacher direction, and record any recovery after force is removed. These distinguish possible causes but do not prove a cause without evidence. No guessed correction or springback allowance.

Written response [p28-forming-response]. Address all three possible causes—marking, setup and material response—and the evidence that could distinguish them. Do not treat the same symptom as proof of one cause.

TEACHER GUIDE

Student page 34 · response guidance

Care for tools and act responsibly

Evidence record [studio-maintenance]. A credible record identifies the real tool/condition/date, current teacher-approved maintenance action/source, actual individual performance and teacher observation, result and evidence/checkpoint. Cleaning/lubrication/consumable replacement must be suitable and authorised; do not supply a generic operating recipe. Mark pending work honestly. A reflection alone cannot certify practical performance.

Evidence record [studio-ethics]. Record a specific real context and the teacher-issued protocol/appropriate authority; explain the student's applied decision and why it follows the protocol; identify evidence, attribution/assistance and actual checkpoint. A hypothetical intention may be labelled for learning but is not completed practical evidence. Do not invent cultural permission or consent.

TEACHER GUIDE

Student page 35 · response guidance

Follow materials through their life

Written response [p35-resource-impact]. The response must explicitly distinguish finite ore from renewable energy, explain plausible environmental and societal effects including Country/Community impacts, and identify whose Country/Community and appropriate authority are involved. It must not invent a consultation or consent. Examples may be conceptual if labelled; stronger responses identify the evidence needed to judge the actual sourcing decision.

TEACHER GUIDE

Student page 36 · response guidance

Compare methods, then record the work

Evidence record [studio-joining]. Record the actual teacher-authorised range and distinct samples, identifying each technique, source/checkpoint, purpose, observed result and evidence ID. Conceptual comparison should address permanence/disassembly, access, material or heat implications as relevant. The source names soldering, welding, fasteners and rivets, but the current permitted practical range and SOPs are teacher-controlled. A blank or pending entry is not proof of performance; sample evidence must exist.

Evidence record [studio-digital]. Require distinguishable actual traditional and digital component or surface-detailing evidence, with the approved method/technology and file/source, student contribution, observed outcome and checkpoint. Files/images need traceable IDs; a concept-only digital sketch does not by itself establish practical creation. No equipment settings or local SOPs are invented; identify pending work honestly.

TEACHER GUIDE

Student page 37 · response guidance

Record two finishing techniques

Source task 7.1 [m07-task01]. One actual surface record must show its initial condition, the confirmed preparation, inspection evidence and what was found before application. Match the source/SOP/product and teacher permission to the real material. A generated image or retrospective invented result is not evidence.

Source task 7.2 [m07-task02]. The justification needs two confirmed criteria, a reason for each and the current product/technical/SDS or teacher source used. Appearance alone is insufficient. Do not infer heat resistance, food-contact suitability, cure or permission from an image or generic coating name.

Evidence record [finish-A]. Technique A must be an actual authorised application with identified process/product/source, before-condition and preparation, teacher observation, quality review and evidence. The last two rows explicitly justify the specified finish against two confirmed criteria. A teacher-approved sample may count; a planned action alone may not. Each field relates to the same real occurrence.

Evidence record [finish-B]. Technique B must be a second distinct appropriate finishing technique with its own authorised source, preparation, actual observed application, quality review and evidence. It cannot merely repeat a stage of Technique A. Record any pending performance explicitly. Together the two records support the program range requirement only when both actually occurred.

TEACHER GUIDE

Student page 38 · response guidance

Connect a decision to its evidence

Written response [p38-transfer]. Name a relevant new or emerging technology, identify supporting evidence and make a reasoned judgement about its career or skill impact. For example, digital inspection may increase demand for interpreting data and maintaining sensors. Link the judgement to an enterprise decision about training, cost, quality or responsibility; identify evidence that could confirm or challenge it. Do not assume automatic superiority, job loss or a proven saving. A general PMI response without a named technology and career/skill judgement is incomplete.

TEACHER GUIDE

Student page 39 · response guidance · 1/2

Connect the brief to proof

Source task 1.1 [m01-task01]. The table must quote or accurately identify at least one confirmed brief requirement, state a suitable observable or measurable proof and name the source. A general intention such as make it well lacks a criterion. Proof remains planned until actually collected.

Written response [p39-purpose]. Identify the real need, intended user and intended function from the confirmed brief. Explain why the requirements serve that function, while flagging any unclear detail for teacher confirmation.

Evidence record [p39-requirements]. Each row links a real requirement and source to a meaningful planned evidence item and checkpoint. A written dimension needs measured evidence; a process requirement needs the relevant record or observation. Do not invent values, pass results or approvals.

Project brief and function [folio-01]. Expect a coherent account of need, intended user, function and teacher-issued requirements. The requirements/evidence table must link them to proof. Accept an explicit unresolved question rather than an invented specification.

Evidence checked and ready [folio-01-complete]. Tick only after the actual response and evidence have been checked for completeness, source accuracy and readability. Pending approvals, measurements or work must be identified. This is a readiness check, not teacher permission or automatic submission.

Optional authorised photo [folio-01-photo]. Optional: accept an authorised photograph or traceable evidence ID of the student's own work when relevant. The caption must explain the feature, check and result. No photo is required by this optional field; do not accept generated illustrative imagery as production evidence.

Evidence caption [folio-01-caption]. A useful caption identifies the student's actual evidence and explains what it shows, why it matters and the requirement or decision it supports. It does not merely name a process or claim an unverified pass.

TEACHER GUIDE

Student page 39 · response guidance · 2/2

Connect the brief to proof

Source or teacher checkpoint [folio-01-source]. Name the actual drawing/view/note, brief, current SOP/WMS, demonstration or feedback and its traceable location/date where available. Record who confirmed an unresolved detail and when; do not invent approval.

TEACHER GUIDE

Student page 40 · response guidance · 1/2

Keep requirements and changes traceable

Evidence record [p40-change]. Distinguish required results from material/time/process or other verified limits. A modification needs its reason, actual approval/source and effect on drawing, list, sequence or evidence. No change is also a valid truthful record.

Source task 9.3 [m09-task03]. A final integrity statement identifies verified physical and documentary evidence, lists actual approved changes or states none, records limitations/unresolved issues, and acknowledges sources and assistance. It must be updated at completion and must not claim a check, approval or submission that did not occur.

Written response [p40-integrity]. Expect a connected first-person account supported by evidence IDs and actual approvals. Cross-check claims with the finished work and records. Explicitly identify pending items or absence of changes rather than inventing a complete result.

Requirements and modifications [folio-02]. The student separates requirements and constraints, then explains only teacher-approved changes. Check the final statement against actual as-built and documentary evidence, sources and assistance.

Evidence checked and ready [folio-02-complete]. Tick only after the actual response and evidence have been checked for completeness, source accuracy and readability. Pending approvals, measurements or work must be identified. This is a readiness check, not teacher permission or automatic submission.

Optional authorised photo [folio-02-photo]. Optional: accept an authorised photograph or traceable evidence ID of the student's own work when relevant. The caption must explain the feature, check and result. No photo is required by this optional field; do not accept generated illustrative imagery as production evidence.

Evidence caption [folio-02-caption]. A useful caption identifies the student's actual evidence and explains what it shows, why it matters and the requirement or decision it supports. It does not merely name a process or claim an unverified pass.

TEACHER GUIDE

Student page 40 · response guidance · 2/2

Keep requirements and changes traceable

Source or teacher checkpoint [folio-02-source]. Name the actual drawing/view/note, brief, current SOP/WMS, demonstration or feedback and its traceable location/date where available. Record who confirmed an unresolved detail and when; do not invent approval.

TEACHER GUIDE

Student page 41 · response guidance · 1/2

Trace one operation and annotate the SOP

Source task 1.2 [m01-task02]. Use a teacher-confirmed operation to state the sequence and project-specific hazards/controls/permission/inspection points. A WMS applies current SOP requirements to the task; copying the SOP word for word does not explain that application.

Source task 4.1 [m04-task01]. Trace a specific written drawing requirement into the related WMS step, relevant control and inspection evidence. The requirement/source, control purpose and recorded result must remain linked to the same real operation.

Evidence record [p41-wms]. Use the same operation for both WMS prompts. Identify the source requirement, WMS step, actual risk judgement with stronger controls before PPE, teacher permission/hold point and inspection record. This short trace supports the full current WMS; it does not authorise work or replace the required document.

Source task 3.2 [m03-task02]. Annotate a genuine teacher-issued SOP excerpt. Identify one actual confirmed control and explain its purpose; identify a distinct actual hold point and explain what must be checked there. A generic list of hazards or an invented procedure does not fulfil this task.

Drawing / annotation response [p41-sop]. The annotation must visibly identify the actual SOP excerpt/source, one control and one hold point with different explanatory notes. For example, explaining the purpose of workholding differs from naming the inspection/permission check before continuing; exact procedure follows current school material.

WHS and risk controls [folio-03]. Record actual hazards, assessed risk, stronger controls, PPE and permission points. Require the full current WMS/SOP where the teacher specifies it; this folio is a trace and annotated evidence record.

Evidence checked and ready [folio-03-complete]. Tick only after the actual response and evidence have been checked for completeness, source accuracy and readability. Pending approvals, measurements or work must be identified. This is a readiness check, not teacher permission or automatic submission.

TEACHER GUIDE

Student page 41 · response guidance · 2/2

Trace one operation and annotate the SOP

Optional authorised photo [folio-03-photo]. Optional: accept an authorised photograph or traceable evidence ID of the student's own work when relevant. The caption must explain the feature, check and result. No photo is required by this optional field; do not accept generated illustrative imagery as production evidence.

Evidence caption [folio-03-caption]. A useful caption identifies the student's actual evidence and explains what it shows, why it matters and the requirement or decision it supports. It does not merely name a process or claim an unverified pass.

Source or teacher checkpoint [folio-03-source]. Name the actual drawing/view/note, brief, current SOP/WMS, demonstration or feedback and its traceable location/date where available. Record who confirmed an unresolved detail and when; do not invent approval.

TEACHER GUIDE

Student page 42 · response guidance · 1/2

Compare materials and justify selection

Source task 2.1 [m02-task01]. The justification must address function, manufacturability, availability and finish using the teacher-specified stock form. Shape names alone are insufficient; connect an actual property or geometry to each decision and cite the confirmed source. Do not substitute an unapproved grade or stock.

Evidence record [p42-materials]. The first row names a pure metal and an alloy and compares relevant composition/property/application evidence, not simply two shapes. The second connects a relevant stock form to manufacture. The third considers a secondary material's possible functional role and limits, with credible evidence and clear conceptual status unless approved in the brief.

Written response [p42-selection]. A connected response covers all four named considerations and identifies the authorised selection/source. For example, a hollow section's form may support a function and joining relationship, but grade/condition and local availability must be confirmed. An attractive finish is not proof of compatibility.

Materials investigation [folio-04]. Compare relevant metals or stock forms with evidence, then justify the actual teacher-specified choice. Keep pure-metal/alloy and secondary-material investigation clearly distinguished from authorised construction choices.

Evidence checked and ready [folio-04-complete]. Tick only after the actual response and evidence have been checked for completeness, source accuracy and readability. Pending approvals, measurements or work must be identified. This is a readiness check, not teacher permission or automatic submission.

Optional authorised photo [folio-04-photo]. Optional: accept an authorised photograph or traceable evidence ID of the student's own work when relevant. The caption must explain the feature, check and result. No photo is required by this optional field; do not accept generated illustrative imagery as production evidence.

Evidence caption [folio-04-caption]. A useful caption identifies the student's actual evidence and explains what it shows, why it matters and the requirement or decision it supports. It does not merely name a process or claim an unverified pass.

TEACHER GUIDE

Student page 42 · response guidance · 2/2

Compare materials and justify selection

Source or teacher checkpoint [folio-04-source]. Name the actual drawing/view/note, brief, current SOP/WMS, demonstration or feedback and its traceable location/date where available. Record who confirmed an unresolved detail and when; do not invent approval.

TEACHER GUIDE

Student page 43 · response guidance · 1/2

Compare two concepts against constraints

Source task 8.3 [m08-task03]. Identify two genuine student concepts and compare both against the same verified constraints. A decision matrix narrative should explain the evidence behind the judgements and why the preferred direction is suitable; it is not simply two popularity scores. Record the teacher approval or remaining question.

Drawing / annotation response [p43-concepts]. Two distinct freehand concepts or accurately referenced student sketches should show the relevant idea and how it addresses the confirmed constraints. Distinguish permitted design choice from fixed original geometry and obtain teacher confirmation for modifications.

Written response [p43-decision]. Expect a concise comparative narrative naming at least two meaningful verified constraints, evidence for each concept and a reasoned choice. A clear unapproved proposal may be discussed honestly, but approval must not be inferred.

Research and ideas [folio-05]. Research and concepts should be the student's own acknowledged work. The chosen direction follows verified requirements and actual approval, with explicit reasons and unresolved details.

Evidence checked and ready [folio-05-complete]. Tick only after the actual response and evidence have been checked for completeness, source accuracy and readability. Pending approvals, measurements or work must be identified. This is a readiness check, not teacher permission or automatic submission.

Optional authorised photo [folio-05-photo]. Optional: accept an authorised photograph or traceable evidence ID of the student's own work when relevant. The caption must explain the feature, check and result. No photo is required by this optional field; do not accept generated illustrative imagery as production evidence.

Evidence caption [folio-05-caption]. A useful caption identifies the student's actual evidence and explains what it shows, why it matters and the requirement or decision it supports. It does not merely name a process or claim an unverified pass.

TEACHER GUIDE

Student page 43 · response guidance · 2/2

Compare two concepts against constraints

Source or teacher checkpoint [folio-05-source]. Name the actual drawing/view/note, brief, current SOP/WMS, demonstration or feedback and its traceable location/date where available. Record who confirmed an unresolved detail and when; do not invent approval.

TEACHER GUIDE

Student page 44 · response guidance · 1/2

Read two features from the drawings

Source task 3.1 [m03-task01]. Use one actual confirmed hole feature. The right annotation panel names the drawing/view, approved datum and written feature information; explain transfer, independent check and evidence before drilling. No hole dimensions or operating setup may be invented.

Source task 8.1 [m08-task01]. Use an actual case development. The left panel identifies a real reference, formed feature and hold point, with annotations explaining their roles. Values/allowances come only from the approved source. Keep this distinct from the hole feature.

Evidence caption [folio-06-caption]. Explain what these specific annotations establish, including the limit that a sketch does not grant practical approval. Refer to both features.

Source or teacher checkpoint [folio-06-source]. Identify the exact drawing/view/issue and actual teacher checkpoint for each feature, or state what remains awaiting confirmation.

Working drawing evidence [folio-06]. Two genuine source-feature annotations connect drawing information to material/dimensions/shape/tolerance/assembly as applicable. Unknown details remain questions. Exact source information, not an image estimate, controls the record.

Drawing / annotation response [p44-two-features]. The left and right fields must show distinct development and hole features. Labels and annotations should answer the relevant exact source tasks. Do not award both from one generic sketch.

Evidence checked and ready [folio-06-complete]. Mark only after checking both annotated features and their source/teacher checkpoint. This is evidence readiness, not practical permission or formal submission.

TEACHER GUIDE

Student page 44 · response guidance · 2/2

Read two features from the drawings

Optional authorised photo [folio-06-photo]. Optional: accept an authorised image or evidence ID of the student's own annotated source drawing. It supports but does not replace the two distinct feature explanations. Generated project imagery is not student construction evidence.

TEACHER GUIDE

Student page 45 · response guidance · 1/2

Reconcile the list, then plan two checks

Source task 2.2 [m02-task02]. Describe a routine that matches part identity, drawing view/label, verified dimensions/material/quantity and spreadsheet row to the actual sketch before cutting. Resolve conflicts with the teacher and update the linked records rather than selecting whichever value seems convenient.

Written response [p45-reconcile]. The response names the linked documents/row and a usable sequence of identity, material/dimension/quantity checks, conflict resolution and update. It also explains one actual or clearly proposed efficiency/waste decision without weakening the approved requirements.

Source task 2.3 [m02-task03]. The checklist names a genuine rail/end-cap stage and includes only confirmed drawing features. Each check states what will be verified and the required evidence or permission before proceeding. Do not invent a rail dimension, tolerance or joining method.

Evidence record [p45-rail]. Require a named rail/end-cap stage, source features, observable checks and evidence/hold-point condition. An unverified generic pass tick is insufficient.

Source task 5.1 [m05-task01]. The plate plan links verified drawing references to the planned stage, hold point and expected evidence. It should make clear what needs checking before further material removal or loss of access, without inventing equipment settings or tolerances.

Evidence record [p45-plate]. Expect a specific plate-work stage and drawing reference, a well-placed pause, named check and appropriate expected evidence. Distinguish planned evidence from an actual result.

Cutting, costing and resources [folio-07]. Summarise the real spreadsheet list with a traceable reference, then explain the waste/efficiency decision. Rail and plate checks must be different named records with verified sources.

TEACHER GUIDE

Student page 45 · response guidance · 2/2

Reconcile the list, then plan two checks

Evidence checked and ready [folio-07-complete]. Tick only after the actual response and evidence have been checked for completeness, source accuracy and readability. Pending approvals, measurements or work must be identified. This is a readiness check, not teacher permission or automatic submission.

Optional authorised photo [folio-07-photo]. Optional: accept an authorised photograph or traceable evidence ID of the student's own work when relevant. The caption must explain the feature, check and result. No photo is required by this optional field; do not accept generated illustrative imagery as production evidence.

Evidence caption [folio-07-caption]. A useful caption identifies the student's actual evidence and explains what it shows, why it matters and the requirement or decision it supports. It does not merely name a process or claim an unverified pass.

Source or teacher checkpoint [folio-07-source]. Name the actual drawing/view/note, brief, current SOP/WMS, demonstration or feedback and its traceable location/date where available. Record who confirmed an unresolved detail and when; do not invent approval.

TEACHER GUIDE

Student page 46 · response guidance · 1/2

Show the sequence and justify the pause

Source task 1.3 [m01-task03]. A verified branch includes logical dependencies, relevant inputs/processes/outputs and an initial quality hold point with a reason. It uses actual source information and shows what happens if the check is unresolved.

Source task 6.2 [m06-task02]. Add one clearly identified, justified hold point to the verified flowchart. Date the revision and state which evidence is reviewed before release. Preserve the distinction between the original justified hold point and the added or revised one.

Drawing / annotation response [p46-flowchart]. Accept a coherent verified branch showing dependencies, checks and feedback. Two distinct annotations identify the initial and later added/revised hold points. No fixed unconfirmed sequence is supplied; approved sequence changes must be dated and sourced.

Written response [p46-reasons]. Explain why each pause belongs at its location, especially before irreversible work or reduced access, and what evidence/teacher permission releases the next stage. The revision must be traceable to a date and source/approval.

Production plan and WMS [folio-08]. The plan sequences verified key processes and records actual demonstrations/approvals/evidence checkpoints. A good flowchart makes dependencies and responses to unresolved quality checks visible.

Evidence checked and ready [folio-08-complete]. Tick only after the actual response and evidence have been checked for completeness, source accuracy and readability. Pending approvals, measurements or work must be identified. This is a readiness check, not teacher permission or automatic submission.

Optional authorised photo [folio-08-photo]. Optional: accept an authorised photograph or traceable evidence ID of the student's own work when relevant. The caption must explain the feature, check and result. No photo is required by this optional field; do not accept generated illustrative imagery as production evidence.

TEACHER GUIDE

Student page 46 · response guidance · 2/2

Show the sequence and justify the pause

Evidence caption [folio-08-caption]. A useful caption identifies the student's actual evidence and explains what it shows, why it matters and the requirement or decision it supports. It does not merely name a process or claim an unverified pass.

Source or teacher checkpoint [folio-08-source]. Name the actual drawing/view/note, brief, current SOP/WMS, demonstration or feedback and its traceable location/date where available. Record who confirmed an unresolved detail and when; do not invent approval.

TEACHER GUIDE

Student page 47 · response guidance · 1/2

Make production evidence explain a decision

Source task 4.2 [m04-task02]. For one real supervised machining stage, record the initial condition, maintained reference, checks and final result. Link observations to evidence IDs and the source drawing/WMS. Planned checks cannot be presented as performed work.

Source task 4.3 [m04-task03]. The same stage may be used, but observation, actual measurement, judgement against the criterion and teacher-confirmed next action must remain distinct. Mark unavailable measurements or unconfirmed actions honestly; do not fabricate acceptance.

Evidence record [p47-machining]. All named elements must appear and be separable in a real supervised machining record. A measurement has a value/unit and method/source where relevant; a judgement uses the approved criterion; next action names the actual teacher confirmation. Require the corresponding genuine evidence, not a generic narrative.

Source task 9.2 [m09-task02]. A coherent paragraph links one actual production decision to its evidence, resulting quality and subsequent evaluation. It explains causation or the supported relationship, without claiming that a photograph proves an unobserved event.

Written response [p47-report]. Accept an evidence-backed paragraph such as a verified datum decision linked to documented checks and the resulting assessed fit, followed by a justified evaluation. Use the student's actual outcome, including faults. Record uncertainty and do not substitute the hypothetical example as personal work.

Production evidence [folio-09]. Photograph captions must identify the student's real process, relevant control and observed result. Evidence IDs should let the teacher locate each photograph and compare the narrative with the real work.

Evidence checked and ready [folio-09-complete]. Tick only after the actual response and evidence have been checked for completeness, source accuracy and readability. Pending approvals, measurements or work must be identified. This is a readiness check, not teacher permission or automatic submission.

TEACHER GUIDE

Student page 47 · response guidance · 2/2

Make production evidence explain a decision

Optional authorised photo [folio-09-photo]. Optional: accept an authorised photograph or traceable evidence ID of the student's own work when relevant. The caption must explain the feature, check and result. No photo is required by this optional field; do not accept generated illustrative imagery as production evidence.

Evidence caption [folio-09-caption]. A useful caption identifies the student's actual evidence and explains what it shows, why it matters and the requirement or decision it supports. It does not merely name a process or claim an unverified pass.

Source or teacher checkpoint [folio-09-source]. Name the actual drawing/view/note, brief, current SOP/WMS, demonstration or feedback and its traceable location/date where available. Record who confirmed an unresolved detail and when; do not invent approval.

TEACHER GUIDE

Student page 48 · response guidance · 1/2

Keep three quality occurrences distinct

Source task 3.3 [m03-task03]. Record an actual drilled feature with its approved criterion, observation/measurement evidence, judged result and next action. A rough rim cannot alone prove size/location or cause; any rework requires the teacher's confirmed action.

Evidence record [p48-drilling]. This record must concern the real drilled feature. The criterion, evidence, result and action must be traceable. It cannot be reused as a different fit-up or assembly occurrence merely by relabelling it.

Source task 5.3 [m05-task03]. Record a real fit-up check: named criterion/source, evidence, issue or supported satisfactory result and teacher-confirmed action. Identify the related components and preserve uncertainty about the cause of any mismatch.

Evidence record [p48-fitup]. Require a distinct fit-up occurrence with identified components, criterion, evidence, observed issue/result and teacher action. Restraint can mask poor contact; unconfirmed acceptance or cause should not be asserted.

Source task 6.3 [m06-task03]. Record a genuine assembly criterion, its evidence, result and follow-up. Relate the entry to the wider assurance process and include the approved recheck after any correction. An unticked or pending follow-up is more accurate than a fictitious pass.

Evidence record [p48-assembly]. Require the actual assembly stage and criterion, evidence, result and follow-up. This third record must not simply repeat the drilling or fit-up occurrence. Confirmed corrections and rechecks should be traceable.

Quality and tolerance checks [folio-10]. Require three actual quality records using teacher-approved criteria, useful measurements or fit evidence, a result and any authorised correction. Never invent tolerances or acceptance.

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Student page 48 · response guidance · 2/2

Keep three quality occurrences distinct

Evidence checked and ready [folio-10-complete]. Tick only after the actual response and evidence have been checked for completeness, source accuracy and readability. Pending approvals, measurements or work must be identified. This is a readiness check, not teacher permission or automatic submission.

Optional authorised photo [folio-10-photo]. Optional: accept an authorised photograph or traceable evidence ID of the student's own work when relevant. The caption must explain the feature, check and result. No photo is required by this optional field; do not accept generated illustrative imagery as production evidence.

Evidence caption [folio-10-caption]. A useful caption identifies the student's actual evidence and explains what it shows, why it matters and the requirement or decision it supports. It does not merely name a process or claim an unverified pass.

Source or teacher checkpoint [folio-10-source]. Name the actual drawing/view/note, brief, current SOP/WMS, demonstration or feedback and its traceable location/date where available. Record who confirmed an unresolved detail and when; do not invent approval.

TEACHER GUIDE

Student page 49 · response guidance · 1/2

Compare processes and explain consistency

Evidence record [p49-industry]. A sound comparison names the same functional process in classroom and industry, then explains a relevant difference in scale, repeatability, control, labour/skills or resource use and a specific societal/environmental consequence. Avoid unsupported claims that all automation improves every outcome.

Source task 5.2 [m05-task02]. Explain how an actual authorised template, common datum or comparison check supports corresponding features and repeatability. Relate it to the demonstrated process and identify the source/permission; a reference itself can repeat error if not verified.

Source task 6.1 [m06-task01]. Explain how consistent references and orientation allow related leg components to be compared before assembly, preventing individual variations from accumulating. Distinguish a mismatch symptom from a confirmed cause; any correction remains teacher-controlled.

Written response [p49-repeatability]. The first explanation links a verified reference/template/comparison to repeatability in the demonstrated grill/guard process. The second links matched leg checks to accumulated variation and assembly stability/alignment. Both must name relevant source evidence and avoid inventing project geometry or correction methods.

Industry and impacts [folio-11]. Compare a genuine classroom and industrial process with evidence, then explain a relevant social or environmental consequence. The two written explanations address different component relationships and cannot be replaced by the general comparison.

Evidence checked and ready [folio-11-complete]. Tick only after the actual response and evidence have been checked for completeness, source accuracy and readability. Pending approvals, measurements or work must be identified. This is a readiness check, not teacher permission or automatic submission.

Optional authorised photo [folio-11-photo]. Optional: accept an authorised photograph or traceable evidence ID of the student's own work when relevant. The caption must explain the feature, check and result. No photo is required by this optional field; do not accept generated illustrative imagery as production evidence.

TEACHER GUIDE

Student page 49 · response guidance · 2/2

Compare processes and explain consistency

Evidence caption [folio-11-caption]. A useful caption identifies the student's actual evidence and explains what it shows, why it matters and the requirement or decision it supports. It does not merely name a process or claim an unverified pass.

Source or teacher checkpoint [folio-11-source]. Name the actual drawing/view/note, brief, current SOP/WMS, demonstration or feedback and its traceable location/date where available. Record who confirmed an unresolved detail and when; do not invent approval.

TEACHER GUIDE

Student page 50 · response guidance · 1/2

Evaluate the evidence and improve the result

Source task 9.1 [m09-task01]. A Plus must cite evidence of a success; a Minus identifies a limitation and supported cause or clearly labelled possible cause; Interesting identifies something that leads to further investigation. Tie all three to confirmed function, quality or process criteria and propose a realistic improvement.

Evidence record [p50-pmi]. Each entry requires an actual observation and evidence. The Minus separates evidence from a proposed cause. Interesting leads to a purposeful check or investigation. The final improvement must connect to a supported issue and feasible teacher-approved action.

Source task 7.3 [m07-task03]. Evaluate one actual confirmed finish result using evidence, a limitation, maintenance implications and an achievable improvement. Appearance alone cannot establish coating identity, cure, adhesion or suitability. Current product information and teacher-approved criteria determine the judgement.

Written response [p50-finish]. A strong response names the actual authorised finish and check, judges the result against the confirmed criterion, acknowledges a limitation and explains a maintenance consequence. The improvement is achievable and does not invent a new coating, unsafe process or unverified service claim.

PMI and final evaluation [folio-12]. The final evaluation judges function, quality and process with genuine evidence and identifies a realistic improvement. Reward honest limitations and supported causal reasoning rather than exaggerated success claims.

Evidence checked and ready [folio-12-complete]. Tick only after the actual response and evidence have been checked for completeness, source accuracy and readability. Pending approvals, measurements or work must be identified. This is a readiness check, not teacher permission or automatic submission.

Optional authorised photo [folio-12-photo]. Optional: accept an authorised photograph or traceable evidence ID of the student's own work when relevant. The caption must explain the feature, check and result. No photo is required by this optional field; do not accept generated illustrative imagery as production evidence.

TEACHER GUIDE

Student page 50 · response guidance · 2/2

Evaluate the evidence and improve the result

Evidence caption [folio-12-caption]. A useful caption identifies the student's actual evidence and explains what it shows, why it matters and the requirement or decision it supports. It does not merely name a process or claim an unverified pass.

Source or teacher checkpoint [folio-12-source]. Name the actual drawing/view/note, brief, current SOP/WMS, demonstration or feedback and its traceable location/date where available. Record who confirmed an unresolved detail and when; do not invent approval.

TEACHER GUIDE

Program coverage and evidence route

Assigned rows 1–6 of 18; not a timetable

Outline the principles of risk management and implement risk-reduction procedures Student pages 7, 41; fields p41-wms. Risk, stronger controls, PPE and permission linked to an actual operation.

Maintain hand tools and equipment including cleaning, lubricating or replacing consumables Student pages 34; fields studio-maintenance. Actual authorised maintenance performance, teacher observation and result.

Apply ethical protocols when making metal projects Student pages 34; fields studio-ethics. Applied teacher-issued ethical protocol and an individual decision record.

Demonstrate safe workshop practices and procedures when using materials, tools and equipment in metal projects Student pages 34; fields studio-ethics. Distinct safe-practice reinforcement belongs to the relevant actual studio occurrence.

Use subject-specific terminology to communicate industrial technology concepts Student pages 7, 41; fields p41-wms. Apply SOP, WMS, hazard, risk, control and hold-point terminology to the operation.

Explain the impact of the metal industry on Country and Aboriginal and Torres Strait Islander Communities Student pages 35; fields p35-resource-impact. Explain Country/Community impacts and identify the appropriate authority question.

TEACHER GUIDE

Program coverage and evidence route

Assigned rows 7–12 of 18; not a timetable

Explain the environmental and societal impact of renewable and non-renewable resources used in the development and production of metal projects Student pages 35; fields p35-resource-impact. Distinguish finite ore and renewable energy; explain social and environmental consequences.

Compare the properties and applications of a range of pure metals and alloys to assess their suitability for making components or metal products Student pages 35, 42; fields p42-materials, p42-selection. Compare pure metal and alloy property/application evidence, then justify the confirmed selection.

Use traditional and digital technologies to create components and detail surfaces for metal products Student pages 36; fields studio-digital. Separate traditional and digital performance records, sources/files, results and checkpoints.

Demonstrate safe workshop practices and procedures when using materials, tools and equipment in metal projects Student pages 36; fields studio-digital. Distinct safe-practice reinforcement belongs to the relevant actual studio occurrence.

Identify and describe secondary materials, such as ceramics, timber, polymers and textiles, that could complement metal-based projects Student pages 35, 42; fields p42-materials. Investigate a secondary material's possible complementary role and constraints.

Use a range of permanent and semi-permanent joining and fabrication techniques, including soldering, welding, fasteners and rivets Student pages 36; fields studio-joining. Distinct actual joining samples with permitted range and evidence; unperformed work remains pending.

TEACHER GUIDE

Program coverage and evidence route

Assigned rows 13–18 of 18; not a timetable

Demonstrate safe workshop practices and procedures when using materials, tools and equipment in metal projects Student pages 36; fields studio-joining. Distinct safe-practice reinforcement belongs to the relevant actual studio occurrence.

Compare permanent and semi-permanent joining techniques and fastenings for metal, and explain their uses Student pages 36; fields studio-joining. Compare purpose, permanence/disassembly or access using source-based technique evidence.

Investigate and apply work and enterprise skills required in the metal industry Student pages 38; fields p38-transfer. Connect a named technology's career/skill impact to an enterprise decision, responsibility and evidence still needed.

Evaluate the impact of new and emerging technologies on careers and professions in the metal industry Student pages 38; fields p38-transfer. Name a new/emerging technology, cite evidence and judge its career/skill impact, acknowledging uncertainty and verification needs.

Use a range of appropriate finishing techniques to enhance or protect a completed metal product Student pages 37; fields finish-A, finish-B. At least two distinct techniques, each with preparation, observed application and quality review.

Demonstrate safe workshop practices and procedures when using materials, tools and equipment in metal projects Student pages 37; fields finish-A, finish-B. Distinct safe-practice reinforcement belongs to the relevant actual studio occurrence.

TEACHER GUIDE

Glossary for discussion

Terms 1–8

Approved criterion. A confirmed requirement used to judge a feature, process or result; the relevant source supplies any value or limit.

Brief. The authorised statement of required outcomes, limits and evidence. It forms the project's production contract.

Constraint. A verified limit on the possible solution, such as permitted resources or processes.

Datum. An identified reference from which related features or measurements are established. Changing it can break the intended relationship.

Development. A flat representation of surfaces intended to form a three-dimensional item. The source controls features, tabs, allowances and dimensions.

Evidence chain. Traceable links connecting a requirement, production decision, observation or measurement, quality result and evaluation.

Fit-up. Temporary positioning of related components to check their relationships before permanent assembly or joining.

Hold point. A planned pause at which specified evidence or permission must be checked before continuing.

TEACHER GUIDE

Glossary for discussion

Terms 9–16

Input / process / output. What enters a stage, what transforms it, and what results. Checks and feedback help control the sequence.

Manufacturability. How a design and its features can be made using the available approved materials, processes, access and controls.

Non-conformance. A result that does not meet its approved criterion. Record the evidence and obtain the authorised next action.

Orientation. The intended direction, face or relationship of a component. Marks and records help preserve it.

PMI. Plus, Minus and Interesting: evidence-backed success, limitation and a meaningful observation leading to investigation.

PPE. Personal protective equipment. It forms part of the required controls but does not replace stronger risk controls or permission.

Quality assurance / control. Assurance is the system that plans, checks and maintains quality; control checks a result at a particular stage.

Repeatability. Producing corresponding features consistently through verified references and controlled methods; a wrong reference can repeat an error.

TEACHER GUIDE

Glossary for discussion

Terms 17–24

RHS. Rectangular hollow section, a stock form with a rectangular hollow cross-section. The name does not specify grade or size.

Risk / hazard. A hazard can cause harm; risk concerns likelihood and consequence in the actual circumstances.

SDS. Safety Data Sheet: product hazard, handling, storage and control information. It has a different role from appearance or a generic finish name.

SOP / WMS. A Standard Operating Procedure defines approved operating requirements. A Work Method Statement applies them to a particular work sequence.

Tolerance. A permitted variation specified by an authoritative source. Do not invent a tolerance from a picture or an impression of fit.

Drying / curing. Drying reduces wetness or tack; curing develops intended properties. A dry-looking surface may not be fully cured.

Pure metal / alloy. A pure metal contains one metallic element; an alloy combines a metal with other elements. Composition and processing affect properties.

Secondary material. A complementary material such as ceramic, timber, polymer or textile considered for a defined role. A possible role is not project approval.

TEACHER GUIDE

Sources and learning media

Source-grounded teaching; current teacher permission governs practical work

BBQ and Case course: <https://stevencowell.github.io/Yr-10-Metal/bbq-case/> . The frozen course contains nine modules, 27 named theory sections, 270 original checks, 27 written evidence tasks and twelve original folio prompts. Their source wording and option order are retained. New connecting prose, response structures and illustrative examples support the same concepts.

Project resource and drawing register: <https://stevencowell.github.io/Yr-10-Metal/bbq-case/project-resource.html> . The original sheets are `bbq-plan.png`, `bbq-case-base.png`, `bbq-case-lid.png` and `bbq-case-handle-components.png` in `assets/plans/bbq-case/`. Corresponding DWG originals remain in the course source. Consult the full source register and current teacher-issued revision if a detail is unclear.

Current teaching program: <https://stevencowell.github.io/Yr-10-Metal/bbq-case/resources/teacher/BBQ-and-Case-60-Period-Teaching-Program-v2.1-Network-Benchmark.pdf> . Only its assigned supplemental concepts and studios are mapped here. Wider-course learning: <https://stevencowell.github.io/Yr-10-Metal/syllabus-learning.html> .

The source video library is available through the course navigation, with definitions at `bbq-case/youtube-library/video-library.data.js`. Use teacher-led viewing to identify the demonstrated concept, connect it to the written source and explain what still requires local permission. A video is not authority for equipment settings, project substitutions or unsupervised operation.

Source freeze: repository commit `7d79a584e70cd4fdc87e345959853941ffc08daf`; course extraction SHA-256 `d68fe1b05b632681121a20e55016ef4bc27f08010e4f66cb6af4522dfd623c66`. The frozen source map records the original files and hashes. A later website update does not silently change this edition.

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Image and drawing credits

Editorial illustrations 1–5

cover (student page 1). Recognise BBQ and shallow case as related products while respecting the original open-frame BBQ form. Original illustrative artwork created for this educational edition. Original AI-generated editorial illustration. Not authenticated student work or approved construction instructions.

stock-forms (student page 10). Compare material forms by observable cross-section and connect form with manufacturing questions. Original illustrative artwork created for this educational edition. Original AI-generated editorial illustration. Not authenticated student work or approved construction instructions.

lid-fit (student page 28). Explain real-base outside measurement as the input for matching lid calculations, separate from the inner opening. Original illustrative artwork created for this educational edition. Original AI-generated editorial illustration. Not authenticated student work or approved construction instructions.

evidence-detail (student page 38). Support a hypothetical PMI judgement with an observable uneven mating edge while leaving cause and tolerance unresolved. Original illustrative artwork created for this educational edition. Original AI-generated editorial illustration. Not authenticated student work or approved construction instructions.

drilled-feature (student page 13). Distinguish an observed hole-rim condition from claims about location, size, cause or acceptance. Original illustrative artwork created for this educational edition. Original illustrative teaching asset, not authenticated student work or approved construction instructions.

TEACHER GUIDE

Image and drawing credits

Editorial illustrations 6–9

turning-surfaces (student page 16). Explain why smooth appearance must be considered alongside size, form, reference surfaces and functional fit. Original illustrative artwork created for this educational edition. Original illustrative teaching asset, not authenticated student work or approved construction instructions.

grill-repeatability (student page 19). Observe repeated spacing and alignment across related elements before evaluating against verified references. Original illustrative artwork created for this educational edition. Original illustrative teaching asset, not authenticated student work or approved construction instructions.

matched-supports (student page 22). Compare related formed pieces from a common reference and record unequal end positions before assembly. Original illustrative artwork created for this educational edition. Original illustrative teaching asset, not authenticated student work or approved construction instructions.

finish-condition (student page 25). Inspect preparation and coverage across faces, bends and edges without inferring coating suitability. Original illustrative artwork created for this educational edition. Original illustrative teaching asset, not authenticated student work or approved construction instructions.

TEACHER GUIDE

Original drawing reproduction

Four complete source sheets, student pages 3–6

Student page 3: original BBQ drawing, assets/plans/bbq-case/bbq-plan.png. Student page 4: case base, bbq-case-base.png. Student page 5: case lid, bbq-case-lid.png. Student page 6: handle components, bbq-case-handle-components.png. These are teacher-course source materials reproduced for the authorised textbook build; original titles, notes, dimensions and source markings remain part of the images.

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